

Michael Yan

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EDUCATION

University of British Columbia

Vancouver, BC

Bachelor of Applied Science in Electrical Engineering (BASc) | Cumulative GPA: 3.7/4.0

2022 – 2026

- Dean's List | Graduated with Distinction

WORK EXPERIENCE

Qingdao Metro-3D | *Electronic Testing, Quality Assurance*

Qingdao, China

Engineering Intern

May 2024 – Aug 2024

- Verified dimensional accuracy across 2 CMM models and 3+ probe configurations using RationalDMIS, performing precision measurements at the micrometer scale to ensure stringent dimensional quality requirements
- Analyzed dimensional measurements and optimized system configurations, reducing measurement discrepancies to within 5–10 μm through repeated probe calibration processes
- Documented dimensional measurements, probe calibration results, and measurement errors, maintaining detailed inspection records for traceability and quality control
- Assembled custom air pressure control systems with electronic pneumatic regulators for air-bearing operations, integrating and configuring hardware for precision equipment

PROJECTS

iHear2++: Acoustic Ear Measurement System (UBC Capstone) | *KiCAD, SolidWorks, Altium, PCBA* 2026

- Developed an acoustic measurement system to calculate personalized ear canal reflectance coefficient across a 20 Hz–20 kHz frequency range for ear health research
- Architected and prototyped 5 MEMS microphone PCBs and a 4-layer STM32H723VGT6 control PCB to synchronously acquire and transfer 5-channel PDM data to a host PC
- Validated signal processing through multi-acquisition averaging, achieving $3.85\times$ noise reduction (11.71 dB) across 3 recordings compared with a theoretical 4.77 dB improvement
- Enhanced signal integrity across hardware interfaces, reducing data-clock crosstalk by approximately 80% through shielded cabling and source-series termination, satisfying the system's sub-1 V interference threshold

Mini Quadcopter (Personal Project, WIP) | *KiCAD, Arduino, PID Control, PCBA*

2026

- Designed and fabricated a custom KiCAD PCB integrating flight-control circuitry, power distribution, and brushless motor interfaces onto a single board
- Implemented a closed-loop PID flight controller, tuning controller gains to improve disturbance rejection
- Documented hardware schematics, PCB architecture, and design specifications to support systematic debugging and iterative hardware revisions

Residential Electrical System Design (Personal Project) | *AutoCAD, Electrical Design*

2026

- Modeled a $\sim 41\text{ m}^2$ residential electrical system, organizing 8 circuits, 5 lighting fixtures, and 8 receptacles into a coordinated electrical floor plan
- Performed circuit-level load and current calculations for lighting, receptacle, and dedicated loads, evaluating approximately 3.0 kVA of conceptual connected load at a nominal 120 V supply
- Developed a coordinated electrical documentation package including floor plans, circuit schematics, panel schedules, load calculations, and electrical legends using standardized CAD drafting practices

Self-Balancing Robot (University Project) | *Arduino, C, PID Control, MATLAB, Simulink*

2025

- Engineered an autonomous two-wheeled self-balancing robot using Arduino Nano 33 BLE, implementing a complementary filter to fuse accelerometer and gyroscope data, mitigating drift for accurate 6-DOF orientation
- Tuned the PID control system in C, achieving $<2\text{ cm}$ oscillation and $\leq 1\text{ cm}$ drift during steady-state operation
- Integrated ultrasonic sensors and Bluetooth Low Energy (BLE) communication to expand system functionality, managing asynchronous data processing for obstacle avoidance and wireless telemetry

SKILLS

Programming Languages: Python, C

Software: KiCAD, Altium Designer, MATLAB, Simulink, SolidWorks, AutoCAD, LTspice, Microsoft Office

Hardware & Technical: PCBA, Soldering, Voltera, Circuit Design/Debugging, Electronic Testing, Verification/Validation, Arduino